5 **(a)** State what is represented by the Avogadro constant.

....

....

.... [1]

(b) A gas is in a sealed container that has a fixed volume. The thermodynamic temperature T of the gas is varied.

Figure 5.1 shows how the pressure p of the gas varies with NT, where N is the number of molecules of the gas.

Figure 5.2 shows how the mean-square speed $\langle c^2 \rangle$ of the molecules of the gas varies with T.

Figure 5.3 shows how the total kinetic energy E_K of all the molecules of the gas varies with T.

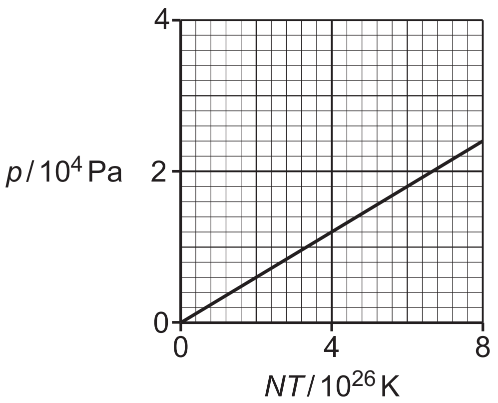

Figure 5.1

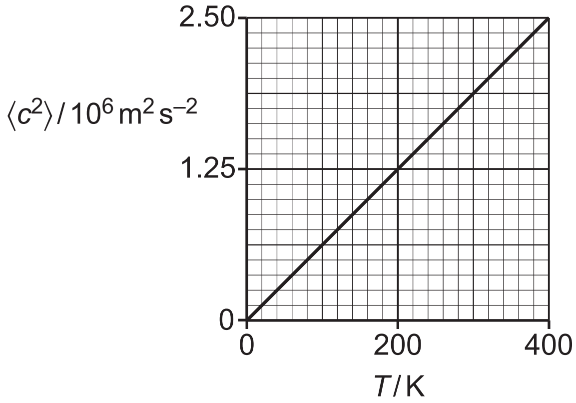

Figure 5.2

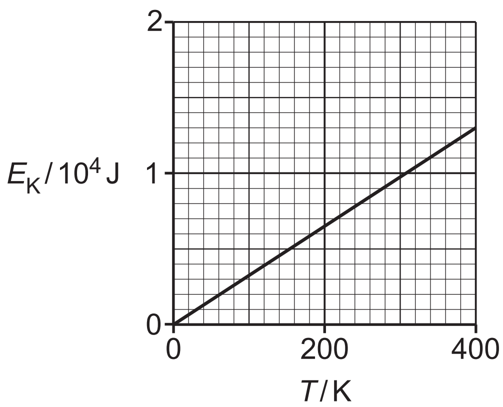

Figure 5.3

Determine **three** conclusions about the gas that can be drawn from Figure 5.1, Figure 5.2 and Figure 5.3. The conclusions may be qualitative or quantitative. Use the space below for any working.

1 ....

....

2 ....

....

3 ....

....
[3]

(c) One assumption of the kinetic theory of gases is that the volume of the molecules is negligible compared with the volume of the gas.

(i) State, with a reason, whether this assumption applies to the gas in **5(b)**.

....

.... [1]

(ii) State **two** other assumptions of the kinetic theory of gases.

1 ....

....

2 ....

....
[2]

[Total: 7]